

Homework 12: Population Scale Analysis

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Table of contents

Population Scale Analysis 1

One sample is not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

Population Scale Analysis

```
# Read the data
url <- "https://bioboot.github.io/bimm143_S26/class-material/rs8067378_ENSG00000172057.6.txt"
expr <- read.table(url)

# Check the data
head(data)
```

```
1 function (... , list = character(), package = NULL, lib.loc = NULL,
2     verbose = getOption("verbose"), envir = .GlobalEnv, overwrite = TRUE)
3 {
4     fileExt <- function(x) {
5         db <- grepl("\\\\.([^.]+\\.)(gz|bz2|xz)$", x)
6         ans <- sub(".*\\.\\.\\.\\.\"", "", x)
```

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

The sample size for A/A is 108 with a median expression level of 31.24847. The sample size for A/G is 233 with a median expression level of 25.06486. The sample size for G/G is 121 with a median expression level of 20.07363.

```
nrow(expr)
```

```
[1] 462
```

Find how many samples belong to each genotype in this dataset

```
# Sample size for each genotype  
table(expr$geno)
```

```
A/A A/G G/G  
108 233 121
```

Find the median expression level for each genotype by using `median()` and extraction `exp` from each genotype group of the dataset

```
# Use median () to find the median for the expression level by genotype group  
median(expr$exp[expr$geno == "A/A"])
```

```
[1] 31.24847
```

```
median(expr$exp[expr$geno == "A/G"])
```

```
[1] 25.06486
```

```
median(expr$exp[expr$geno == "G/G"])
```

```
[1] 20.07363
```

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

Use `ggplot2` to create a boxplot of genotype and expression to compare expression levels by genotype.

```
# Load ggplot2
library(ggplot2)
```

From the boxplot, it is clear that A/A has higher levels of expression than G/G genotype. This suggests that allele variation has an impact/effect or correlation with the expression of ORM DL3. Due to varying expression levels with G/G having the lowest, and A/G having mid-level expression, it can be concluded that the SNP does affect the expression of ORM DL3.

Make a boxplot

```
# Use ggplot to make the boxplot, filling in boxes by color according to genotype
# Set notch=true to define a median for each genotype group
ggplot(expr) + aes(geno, exp, fill=geno) +
  geom_boxplot(notch=TRUE)
```

